Ethical Open Science

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Learning Objectives

1. Think and discuss what open science means
2. Understand the basics of FAIR and CARE
3. Learn where can data can be archived and while considering FAIR and CARE principles

Doing Ethical Open Science

Ethical Open Science for Past Global Change Data RCN

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Breakout session

Discuss	Discuss questions in small groups – 10 minutes
Answer	Write answers – 5 minutes
Read	Discuss answers with other groups – 10 minutes

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Breakout session questions

1. Why is open science important?

2. What information is important to archive in a findable, accessible, interoperable, and reproducible way?

3. What are some repositories where you can archive open science data?

4. What are some ethical considerations with open science data?

Why open science?

1. Transparency
2. Reproducibility
3. Accessibility
4. Collaboration
5. Equity
6. Efficient workflows
7. Education and outreach

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F A I R

Findable

Accessible

Interoperable

Reusable

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What is **FAIR?**

It is a set of principles

Findable

- F1. (Meta)data are assigned a globally unique and persistent identifier
- F2. Data are described with rich metadata (defined by R1 below)
- F3. Metadata clearly and explicitly include the identifier of the data they describe
- F4. (Meta)data are registered or indexed in a searchable resource

Accessible

- A1. (Meta)data are retrievable by their identifier using a standardised communications protocol
  - A1.1 The protocol is open, free, and universally implementable
  - A1.2 The protocol allows for an authentication and authorisation procedure, where necessary
- A2. Metadata are accessible, even when the data are no longer available

Interoperable

- I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (Meta)data use vocabularies that follow FAIR principles
- I3. (Meta)data include qualified references to other (meta)data

Reusable

- R1. (Meta)data are richly described with a plurality of accurate and relevant attributes
  - R1.1. (Meta)data are released with a clear and accessible data usage license
  - R1.2. (Meta)data are associated with detailed provenance
  - R1.3. (Meta)data meet domain-relevant community standards

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DATA PRINCIPLES

INDIGENOUS				MAINSTREAM		
New Zealand Indigenous Data Sovereignty Principles	**Australia Indigenous Data Sovereignty Protocols**	**United States Indigenous Data Governance Principles**	**Canada Indigenous Data Governance Principles**	**Open Data Charter Principles**	**FAIR Principles for Data Management and Stewardship**	**STREAM Properties for Industrial and Commoditized Data**
Authority	Self-Determination	Inherent Sovereignty	OCAP®	Open By Default	Findable	Sovereign
Relationships	Available and Accessible	Indigenous Knowledge	Indigenous Knowledge	Timely and Comprehensive	Accessible	Trusted
Obligations	Collective Rights and Interests	Ethics	Methodology and Approaches	Accessible and Usable	Interoperable	Reusable
Collective Benefit	Accountability	Intergenerational Collective Wellbeing	Evidence to Build Policy	Comparable and Interoperable	Reusable	Exchangeable
Reciprocity	Exercise Control	Relationships	Ethical Relationships	For Improved Governance & Citizen Engagement		Actionable
Guardianship			Data Governance	For Inclusive Development and Innovation		Measurable

People oriented principles

Purpose oriented principles

Data oriented principles

Carroll, S, et al. 2020. The CARE Principles for Indigenous Data Governance. *Data Science Journal*

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Where to archive?

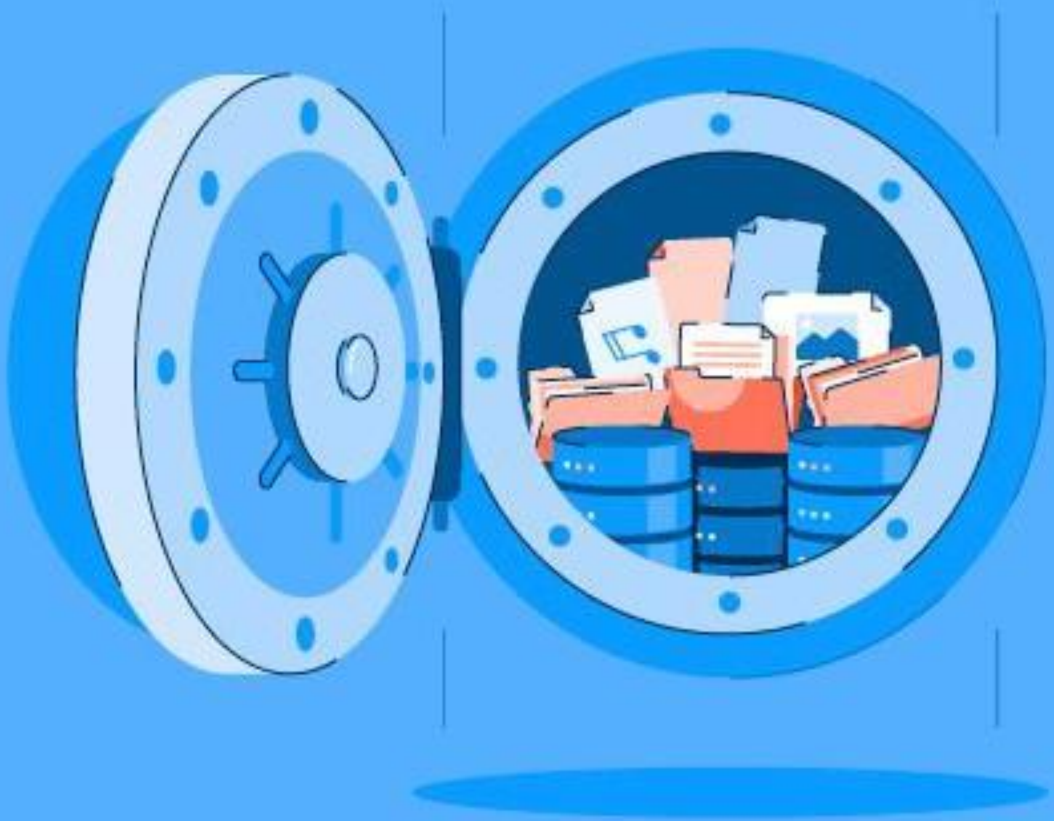
Dryad, Zenodo, FigShare, the Open Science Framework (OSF), Pangaea, institutional and funding body repositories…

General data: PBDB and Neotoma (occurrence data), MorphoSource, MorphoBank…

Where to archive?

But to consider:

- Longevity
- Cost
- Capacity
- Accessibility
- Discoverability
- Citability
- Version control
- Licensing



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